

Instruction Set Principles

E. Sanchez

**Politecnico di Torino
Dipartimento di Automatica e Informatica**

Instruction Set Architecture

The *Instruction Set Architecture* (ISA) is how the computer is seen by the programmer or the compiler.

There are many alternatives possible for the ISA designer, any choice modifies the way to encode the instructions.

The different alternatives may be evaluated in terms of

- Processor performance
- Processor complexity
- Compiler complexity
- Code size
- Power consumption, ...

Different product areas may assign different weights to the above parameters.

ISA Outline

- **Memory access**
- **Addressing modes**
- **ISA operations**
- **Type of operands and operand sizes**
- **Instructions encoding.**

GPR machines

Currently, most of the processors are *General-Purpose Register* (GPR) machines.

The reason for this is:

- the quantity of register is high, usually more than 16
- registers are faster than memory
- registers are easier for a compiler to use.

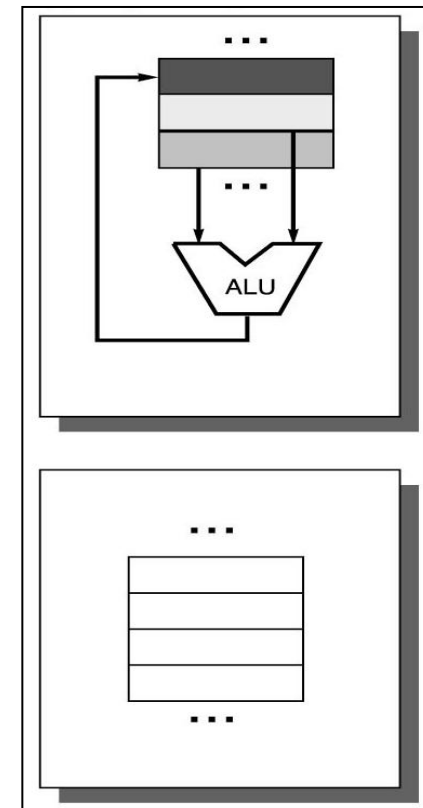
Load-Store Architecture

Today, the most common architecture is Register-Register or Load-Store architecture.

- Mathematical operations only include registers as operands. The data memory is only accessed through *Load* and *Store* instructions.

CPU's can be classified according to

- typical number of operands per ALU instruction (2 or 3)
- typical number of memory operands per ALU instruction (from 0 to 2), no real cases use 3 memory accesses during one computation.



Load-Store

Memory Addresses

Alternatives:

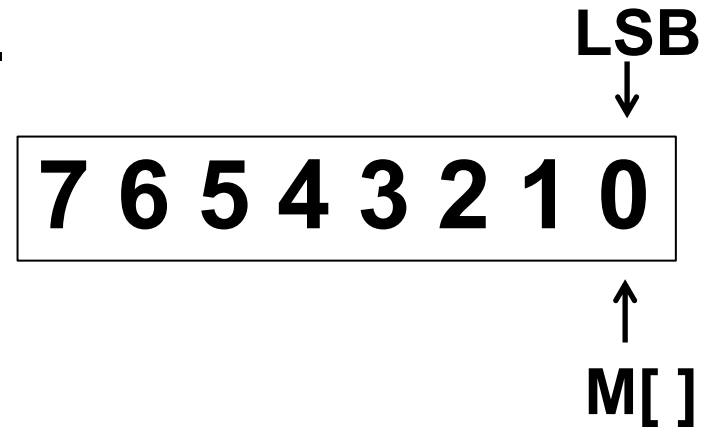
- Little Endian vs. Big Endian
- Aligned vs. misaligned accesses.

Memory Addresses

Alternatives:

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It puts the byte with the lower address (X...X000) at the least significant position.
The address of the data is that of the least significant byte.



Memory Addresses

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LSB
↓

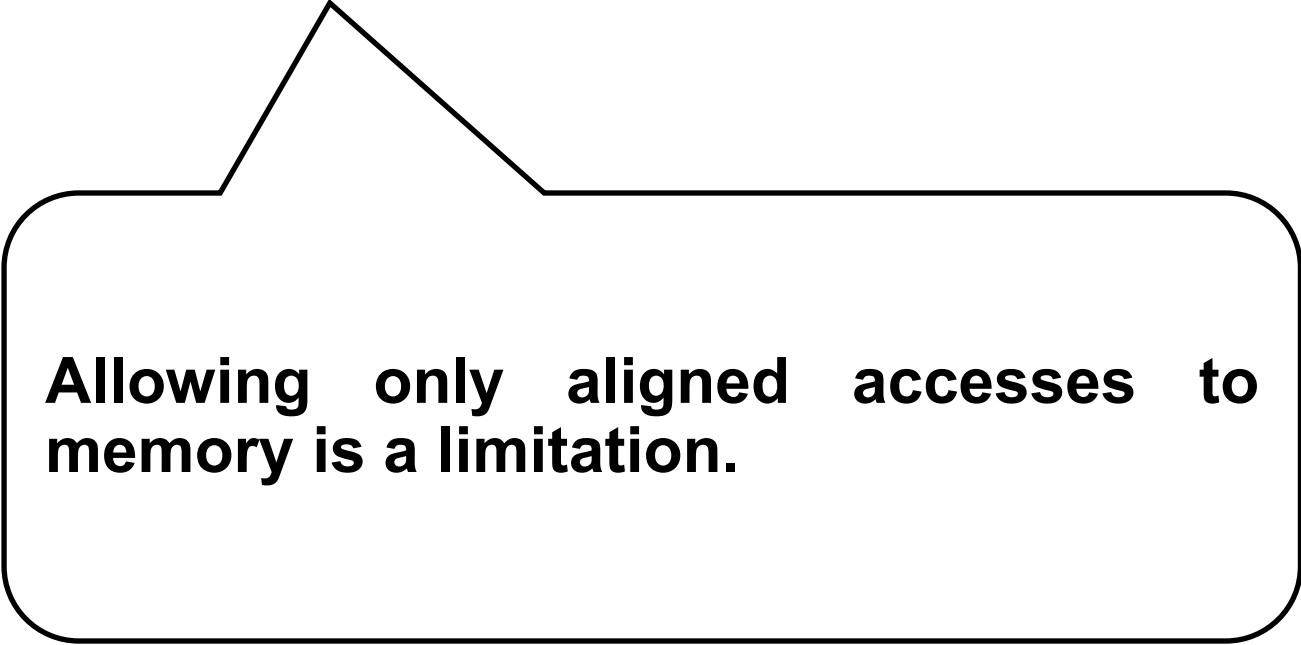
0	1	2	3	4	5	6	7
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↑
M[]

Memory Addresses

Alternatives:

- Little Endian vs. Big Endian
- Aligned vs. misaligned accesses.

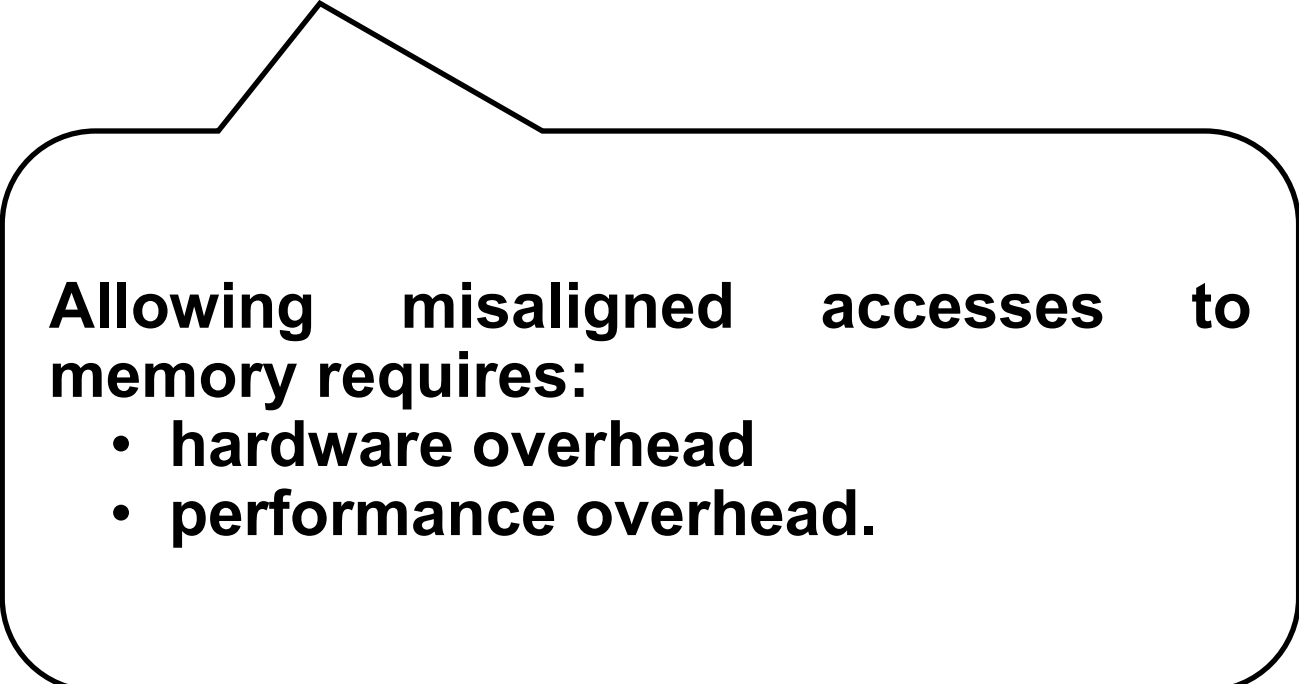


Allowing only aligned accesses to memory is a limitation.

Memory Addresses

Alternatives:

- Little Endian vs. Big Endian
- Aligned vs. misaligned accesses.



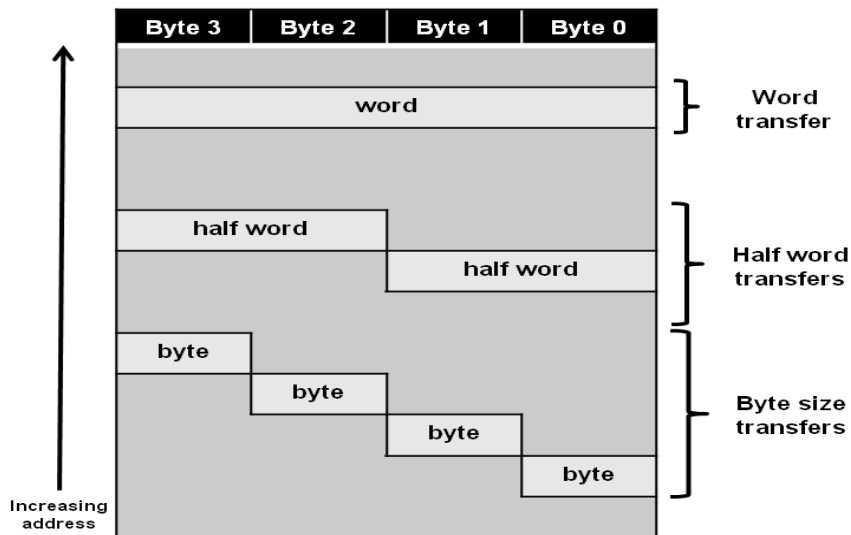
Allowing misaligned accesses to memory requires:

- hardware overhead
- performance overhead.

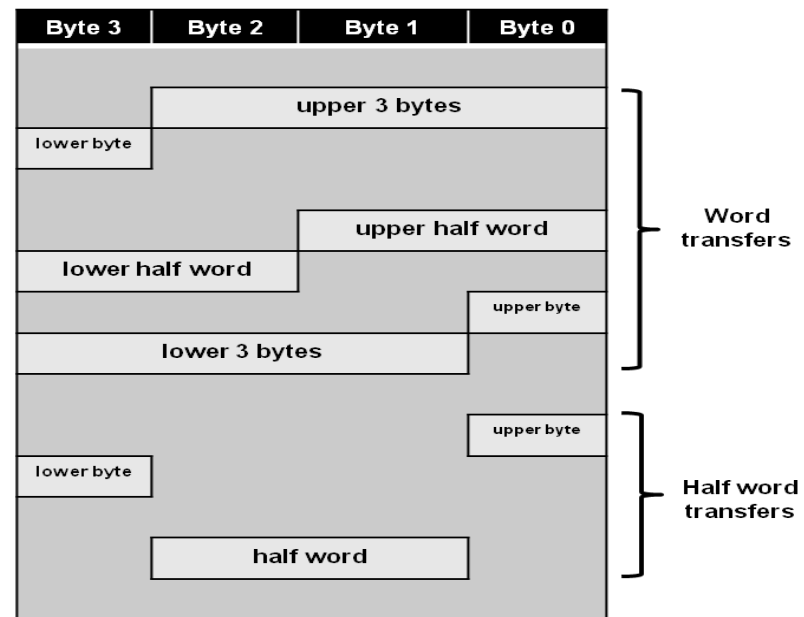
Memory Addresses

Alternatives:

- Little Endian vs. Big Endian
- Aligned vs. misaligned accesses.



Aligned access



Misaligned access

<https://microchip.wdfiles.com/>

Addressing modes

In GPR machines an addressing mode specifies a constant, a register, or a memory location (through its *effective address*).

Register Mode

Addressing mode	Example instruction	Meaning	When used
Register	Add R4, R3	$\text{Regs}[\text{R4}] \leftarrow \text{Regs}[\text{R4}] + \text{Regs}[\text{R3}]$	When a value is in a register.

Immediate Mode

Immediate

Add R4, #3

Regs[R4] ← Regs[R4] + 3

For constants.

Displacement Mode

Displacement	Add R4, 100(R1)	Regs[R4] ← Regs[R4] + Mem[100 + Regs[R1]]	Accessing local variables.
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Register Deferred (or Indirect) Mode

Register deferred
or indirect

Add R4, (R1)

$\text{Regs}[R4] \leftarrow \text{Regs}[R4] +$
 $\text{Mem}[\text{Regs}[R1]]$

Accessing using a pointer or a
computed address.

Indexed Mode

Indexed	Add R3, (R1 + R2)	$\text{Regs}[\text{R3}] \leftarrow \text{Regs}[\text{R3}] + \text{Mem}[\text{Regs}[\text{R1}] + \text{Regs}[\text{R2}]]$	Sometimes useful in array addressing: R1 = base of array; R2 = index amount.
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Direct (or Absolute) Mode

Direct or absolute	Add R1, (1001)	$\text{Regs}[R1] \leftarrow \text{Regs}[R1] + \text{Mem}[1001]$	Sometimes useful for accessing static data; address constant may need to be large.
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Memory Indirect Mode

Memory indirect
or memory
deferred

Add R1, @ (R3)

$\text{Regs}[R1] \leftarrow \text{Regs}[R1] +$
 $\text{Mem}[\text{Mem}[\text{Regs}[R3]]]$

If R3 is the address of a pointer
 p , then mode yields $*p$.

(post) autoincrement Mode

Autoincrement

Add R1, (R2) +

$\text{Regs}[R1] \leftarrow \text{Regs}[R1] +$
 $\text{Mem}[\text{Regs}[R2]]$
 $\text{Regs}[R2] \leftarrow \text{Regs}[R2] + d$

Useful for stepping through arrays within a loop. R2 points to start of array; each reference increments R2 by size of an element, d .

(pre) autodecrement Mode

Auto-
decrement

Add R1, -(R2)

$\text{Regs}[R2] \leftarrow \text{Regs}[R2] - d$
 $\text{Regs}[R1] \leftarrow \text{Regs}[R1] +$
 $\text{Mem}[\text{Regs}[R2]]$

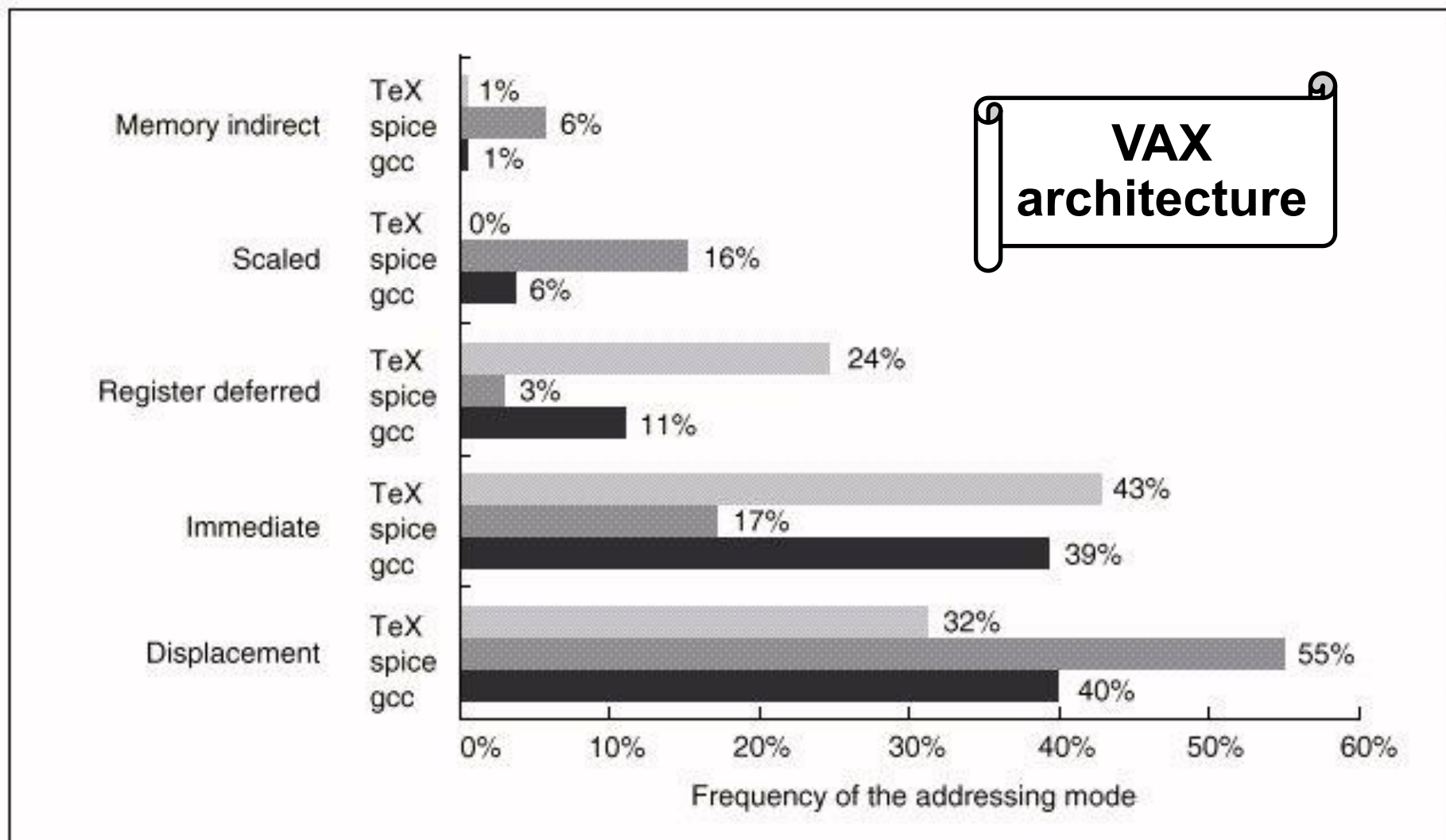
Same use as autoincrement.
Autodecrement/increment can
also act as push/ pop to imple-
ment a stack.

Choosing the addressing modes

By carefully choosing the addressing modes, one can obtain some important consequences:

- **Reducing the number of instructions**
- **Increasing the CPU architecture complexity**
- **Increasing the average Cycles Per Instruction (CPI).**

Usage of addressing modes



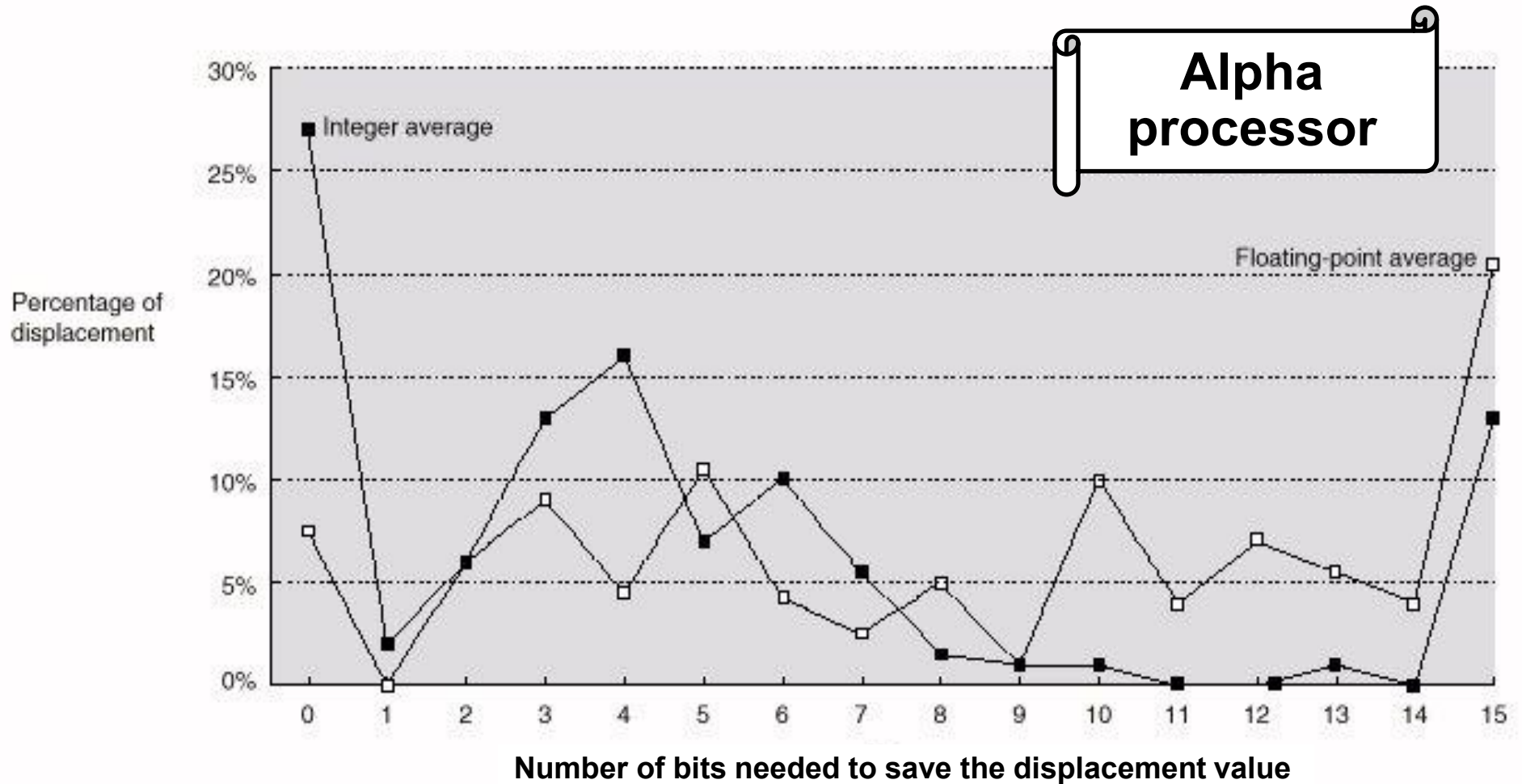
Open issue

When the selected addressing requires a displacement, how many bits should be devoted to it in the instruction code?

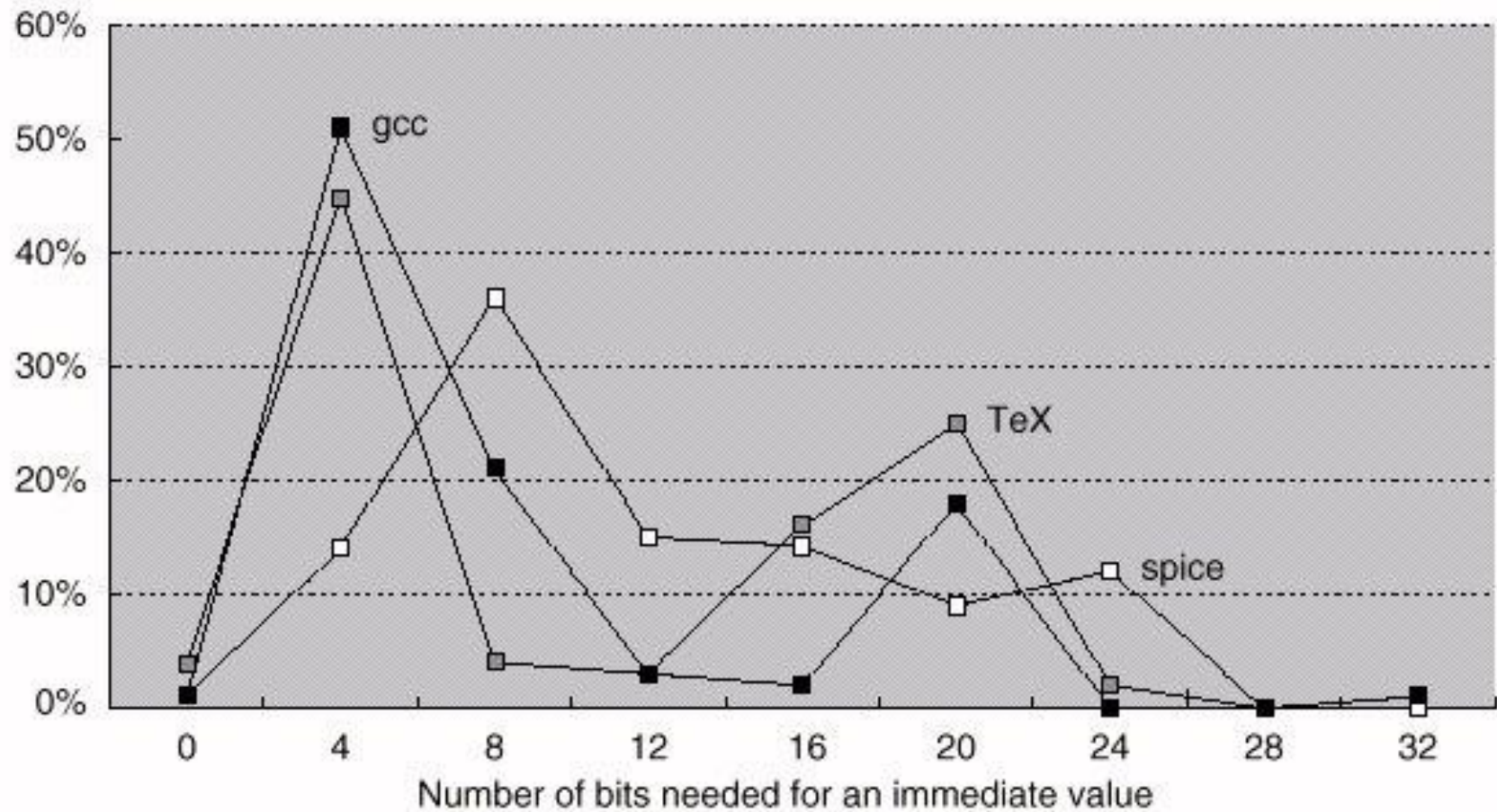
How large can be the immediate value, which is embedded in the addressing part of the instruction?

An experimental evaluation can be performed.

Displacement values



Immediate values



Summary

- **Displacement, immediate and register indirect modes represent from 75% to 99% of the addressing modes**
- **The address size for displacement mode should be from 12 to 16 bits (75% to 99% of the displacements)**
- **The size of the immediate field should be at least 8 or 16 bits (50% and 80% of the cases, respectively).**

Operations in the Instruction Set

Operator type	Examples
Arithmetic and logical	Integer arithmetic and logical operations: add, and, subtract, or
Data transfer	Loads-stores (move instructions on machines with memory addressing)
Control	Branch, jump, procedure call and return, traps
System	Operating system call, virtual memory management instructions
Floating point	Floating-point operations: add, multiply
Decimal	Decimal add, decimal multiply, decimal-to-character conversions
String	String move, string compare, string search
Graphics	Pixel operations, compression/decompression operations

80x86 instruction frequency

Rank	80x86 instruction	Integer average (% total executed)
1	load	22%
2	conditional branch	20%
3	compare	16%
4	store	12%
5	add	8%
6	and	6%
7	sub	5%
8	move register-register	4%
9	call	1%
10	return	1%
Total		96%

RISC-V instruction frequency

Program	Loads	Stores	Branches	Jumps	ALU operations
astar	28%	6%	18%	2%	46%
bzip	20%	7%	11%	1%	54%
gcc	17%	23%	20%	4%	36%
gobmk	21%	12%	14%	2%	50%
h264ref	33%	14%	5%	2%	45%
hmmer	28%	9%	17%	0%	46%
libquantum	16%	6%	29%	0%	48%
mcf	35%	11%	24%	1%	29%
omnetpp	23%	15%	17%	7%	31%
perlbench	25%	14%	15%	7%	39%
sjeng	19%	7%	15%	3%	56%
xalancbmk	30%	8%	27%	3%	31%
AVG≈ 24,58% 11,00% 17,67% 2,67% 42,58%					

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Control Flow Instructions

They can be distinguished in four categories:

- **conditional branches**
- **jumps**
- **procedure calls**
- **procedure returns.**

Conditional branches are by far the most frequently executed control flow instructions.

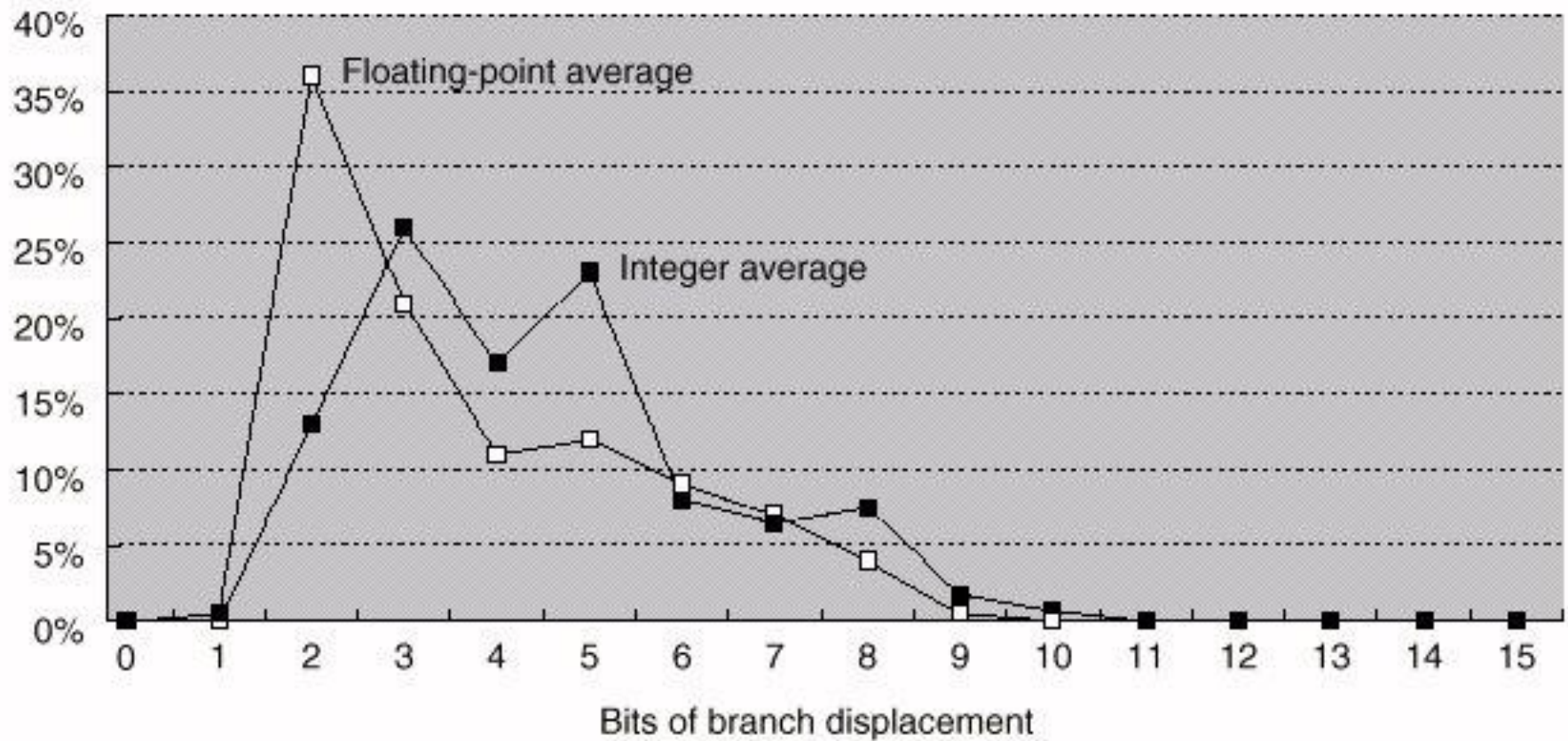
The destination address

It is normally specified as a displacement with respect to the current value of the Program Counter.

In this way:

- **we save bits, since the target instruction is often close to the source one**
- **the code is position-independent.**

Branch distances



Register Indirect Jumps and Procedure Calls

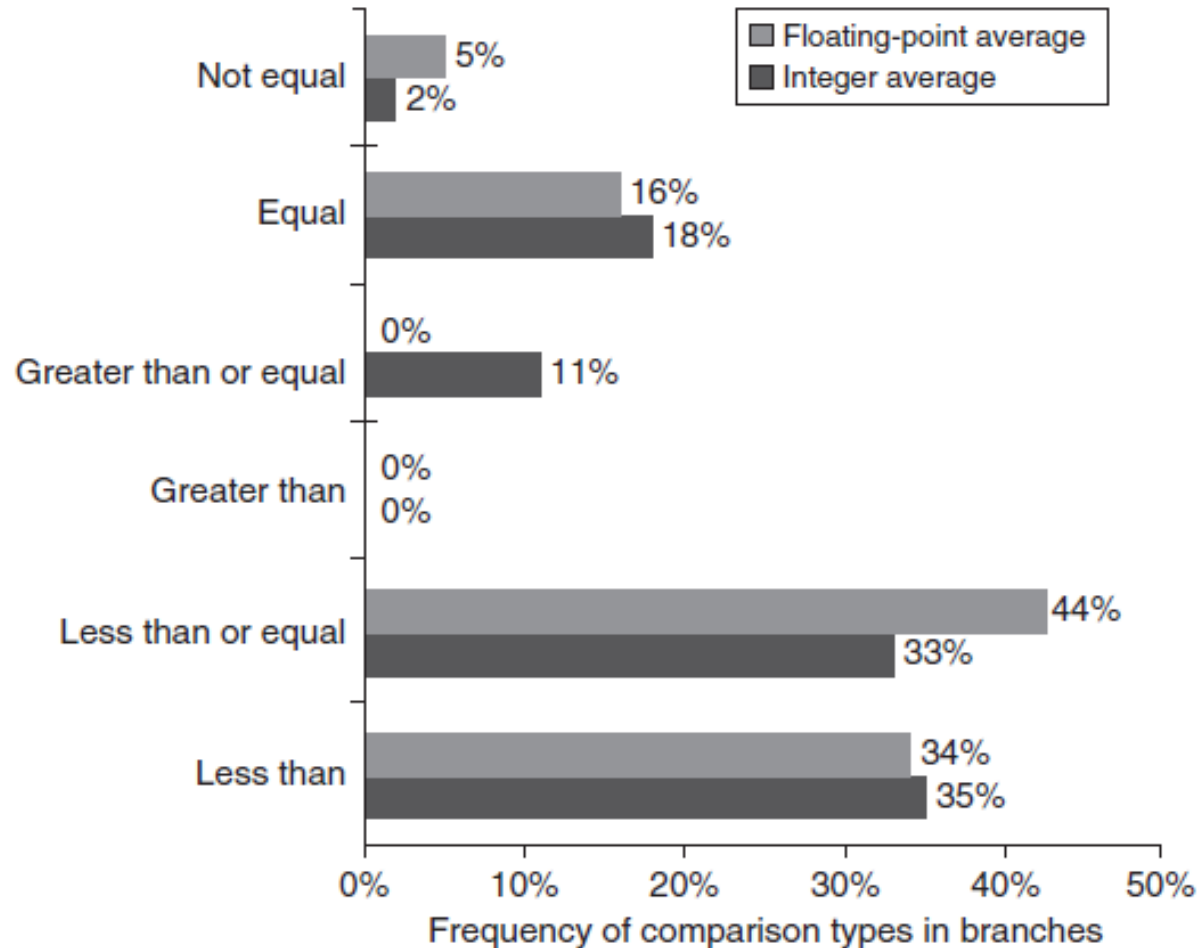
They allow:

- to write code including jumps whose target is not known at compile time
- to implement *case* or *switch* statements
- to support *dynamically shared libraries* (i.e., libraries which are loaded only when called)
- to support *virtual functions* (i.e., calling different functions depending on the data type).

Evaluating Branch Conditions

Name	Examples	How condition is tested	Advantages	Disadvantages
Condition code (CC)	80x86, ARM, PowerPC, SPARC, SuperH	Tests special bits set by ALU operations, possibly under program control	Sometimes condition is set for free.	CC is extra state. Condition codes constrain the ordering of instructions because they pass information from one instruction to a branch
Condition register/ limited comparison	Alpha, MIPS	Tests arbitrary register with the result of a simple comparison (equality or zero tests)	Simple	Limited compare may affect critical path or require extra comparison for general condition
Compare and branch	PA-RISC, VAX, RISC-V	Compare is part of the branch. Fairly general compares are allowed (greater then, less then)	One instruction rather than two for a branch	May set critical path for branch instructions

Conditional branches



Procedures

Some information need to be saved:

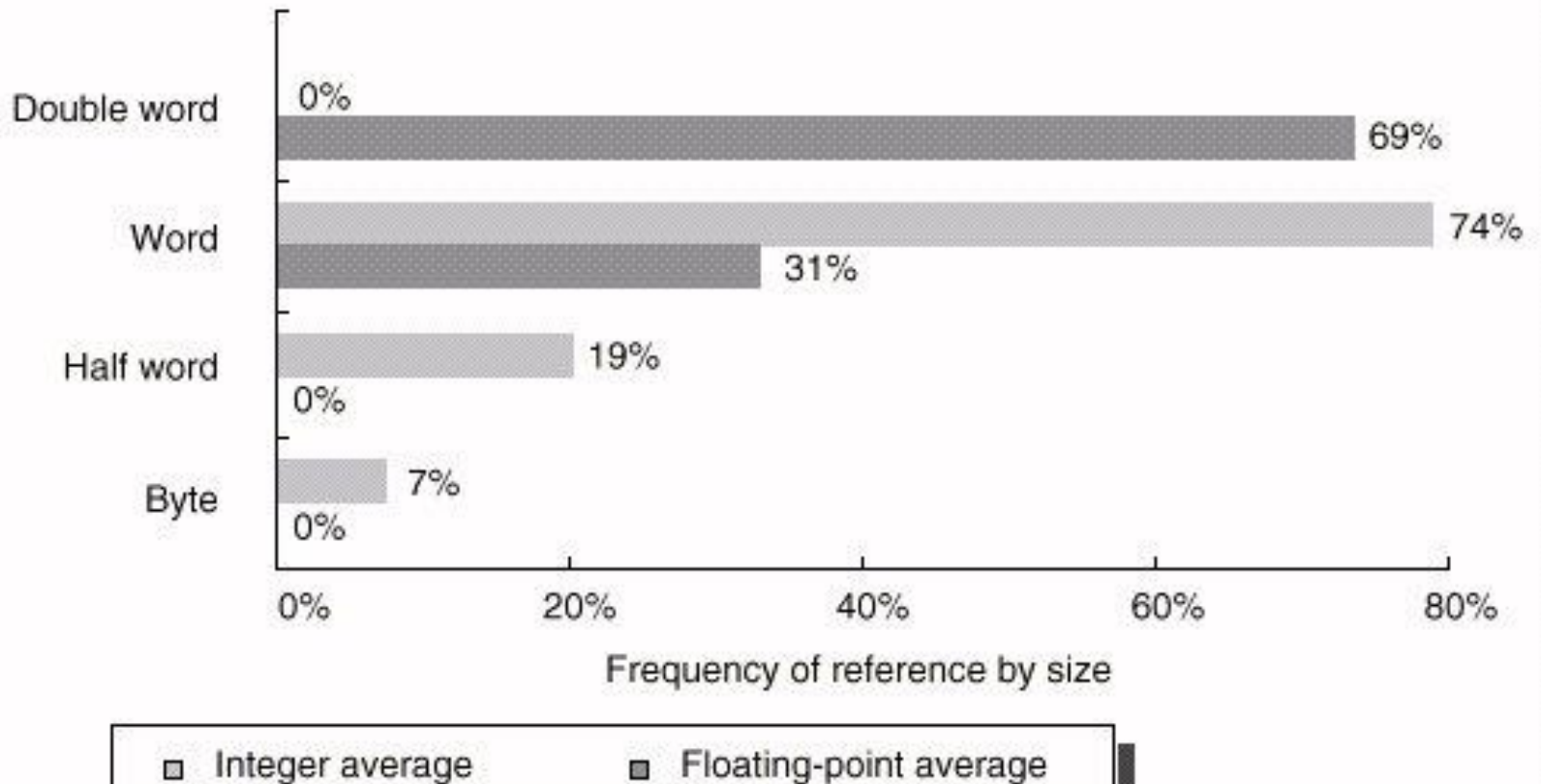
- **the return address**
- **the accessed registers:**
 - **caller saving**
 - **callee saving.**

Type and Size of Operands

Most frequently supported data types:

- **char (1 byte)**
- **half word (2 bytes)**
- **word (4 bytes)**
- **double word (8 bytes)**
- **single-precision floating-point (4 bytes)**
- **double-precision floating-point (8 bytes).**

Distribution of Data Accesses by Size



Instruction Set Encoding

Instruction Set Encoding depends on

- which instructions compose the Instruction Set
- which addressing modes are supported.

When a high number of addressing modes is supported, an *address specifier* field is used to specify the addressing mode and the registers which are possibly involved.

When the number of addressing modes is low, they can be encoded together with the opcode.

Instruction Set Encoding

Operation and no. of operands	Address specifier 1	Address field 1	...	Address specifier n	Address field n
----------------------------------	------------------------	--------------------	-----	--------------------------	----------------------

(A) Variable (e.g., Intel 80x86, VAX)

Operation	Address field 1	Address field 2	Address field 3
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(B) Fixed (e.g., RISC V, ARM, MIPS, PowerPC, SPARC)

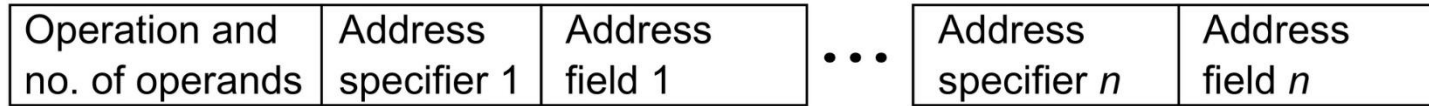
Operation	Address specifier	Address field
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Operation	Address specifier 1	Address specifier 2	Address field
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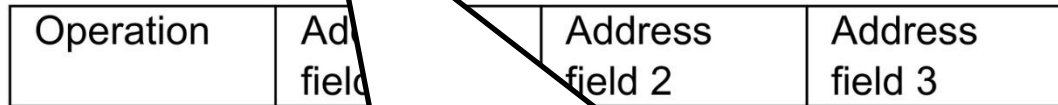
Operation	Address specifier	Address field 1	Address field 2
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(C) Hybrid (e.g., RISC V Compressed (RV32IC), IBM 360/370, microMIPS, Arm Thumb2)

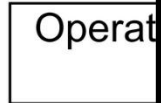
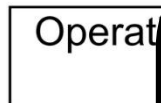
Instruction Set Encoding



(A) Variable (e.g., Intel 80x86, VAX)



(B) Fixed (e.g., RISC, MIPS, PowerPC, SPARC)



(C) Hybrid

- Supports any number of operands
- Instructions have a variable length
- Lower performance
- Minimum code size

(b2)

Instruction Set Encoding

Operation and no. of operands	Address specifier 1	Address field 1	...	Address specifier n	Address field n
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(B) Fixed (e.g., RISC V, ARM, MIPS, PowerPC, SPARC)

Operation	Address specifier	Address field
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Op

Op

(C) R

- Fixed number of operands
- Address specifier included in the opcode
- Fixed instruction length
- Maximum performance
- Larger code size

Instruction Set Encoding

Operation and no. of oper	...	...	...	...
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(A) Variable

Operation

(B) Fixed (

- **Multiple formats (specified by the opcode)**
- **Allow trading-off between code size and performance**

Operation	Address specifier	Address field
-----------	----------------------	------------------

Operation	Address specifier 1	Address specifier 2	Address field
-----------	------------------------	------------------------	------------------

Operation	Address specifier	Address field 1	Address field 2
-----------	----------------------	--------------------	--------------------

(C) Hybrid (e.g., RISC V Compressed (RV32IC), IBM 360/370, microMIPS, Arm Thumb2)

Conflicting Issues

The designer should address several conflicting issues:

- **the code size**
- **the size of the Instruction Set, the number of addressing modes, and the number of registers**
- **the complexity of the fetch and decoding hardware.**